

Electrical characterization

Each populated coupon was characterized electrically to establish which assembly condition produces the better bond. Two rounds were run. The first screened every addressable channel with a handheld DMM to classify it as working or failed and to record how it failed. The second swept current through each surviving channel and fitted the diode equation to extract series resistance, which contains the bond. The screening separates the eight conditions clearly. The series resistance does not, and Section D shows why.

Table I. Measurement setup.

Quantity	Test structure	Instrument	Excitation
Channel state and failure mode	D1–D8 gold probe pads	Handheld DMM, diode test	≈ 1 mA forward
Forward voltage, series resistance	D1–D8, south 32-pin header	Arduino UNO R3, six-branch binary current source, 14-bit oversampled ADC	Pulsed DC, 63 levels, 0.5–15.9 mA, 5 ms on / 250 ms off
Sense-resistor calibration	R_EIS_LOAD, 100 Ω 0.1 %	Handheld DMM, 600 Ω range, REL	DC
Substrate temperature	NTC pads TH1–TH4	Handheld DMM, resistance	DC

A. Channel screening and yield

All 120 addressable channels on the five coupons were probed at the gold pads with a DMM diode test. A channel passes if it shows a forward junction voltage and lights. Failures were classified as open (no conduction), short (junction voltage below 0.15 V), cross-lit (driving one colour lights another, which indicates a bridge between cathodes), detached (the die is physically absent), or suspect.

Yield separates the conditions. A chi-square test on the 120 channels gives $p = 2.2 \times 10^{-4}$. Conditions 5 and 7 lost no channels at all; condition 3 lost 9 of 12.

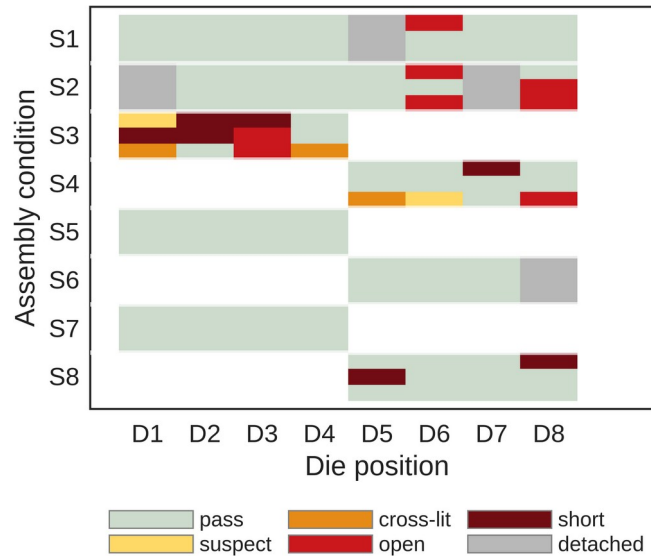


Fig. 1. Channel-level screening result. Three rows per condition (red, green, blue), one column per die position. Blank cells are die positions the condition does not own.

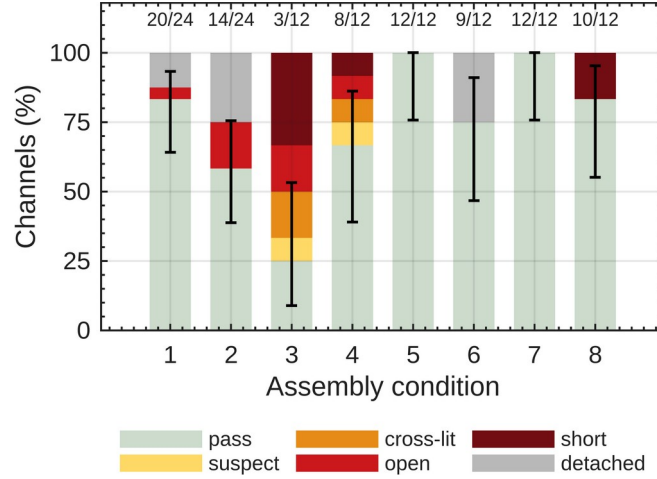


Fig. 2. Yield and failure mode by condition. Error bars are 95 % Wilson intervals on the pass fraction; counts above each bar are passing channels over total.

Table II. Channel yield and failure mode by assembly condition.

Condition	Pass / total	Yield (%)	Failure modes	Interpretation
1	20 / 24	83.3	open 1, detached 3	under-bonded
2	14 / 24	58.3	open 4, detached 6	under-bonded
3	3 / 12	25.0	suspect 1, cross-lit 2, open 2, short 4	over-bonded
4	8 / 12	66.7	suspect 1, cross-lit 1, open 1, short 1	over-bonded
5	12 / 12	100.0	none	in window
6	9 / 12	75.0	detached 3	under-bonded
7	12 / 12	100.0	none	in window
8	10 / 12	83.3	short 2	over-bonded

The failure mode carries more information than the yield number. Conditions 3, 4 and 8 fail through shorts and cross-lit pairs, which is what excess solder and pad-to-pad bridging produce. Conditions 1, 2 and 6 fail the other way, through detached dice and open cathodes, which is the signature of insufficient wetting or a weak joint. Conditions 5 and 7 show neither. On this evidence 5 and 7 sit inside the process window and the other six sit on one side of it or the other.

B. Forward characteristics and series resistance

Each surviving channel was swept over 63 current levels between 0.5 and 15.9 mA. Current comes from six binary-weighted resistors on digital pins, giving 2^6-1 distinct levels, and is pulsed 5 ms on and 250 ms off so every point carries the same thermal history. Series resistance follows from a three-parameter least-squares fit over all points above 0.5 mA:

$$V(I) = V_0 + n \cdot V_T \cdot \ln(I) + I \cdot R_s$$

R_s is the sum of the die's internal resistance, both bonds, and the board traces inside the sense loop. All dice come from one reel, so a shift in mean R_s between conditions would indicate a shift in bond resistance.

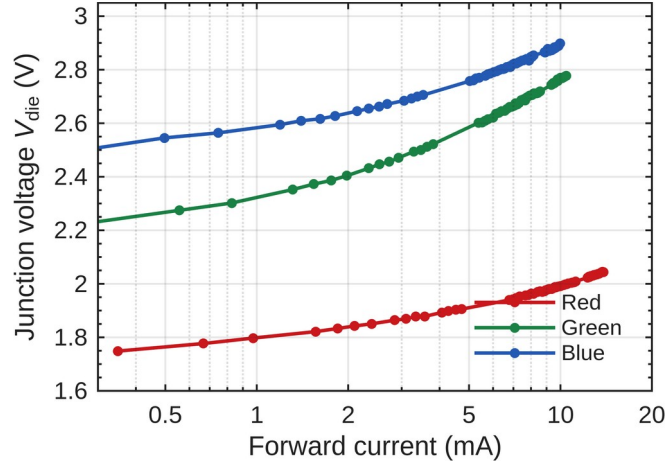


Fig. 3. Forward characteristics of one die (S1 D1) in all three colours. Red reaches 13.9 mA; green and blue stop near 10 mA because the 5 V rail leaves less headroom.

Only the red channels give physical fits. Green and blue return ideality factors between 3.2 and 4.0, which no LED has. At a forward voltage near 2.8 V the 5 V supply allows only a 20-fold current range, over which V_0 , n and R_s trade off against one another and the fit becomes degenerate. Red spans 30-fold and returns ideality 1.60 to 2.06. Everything below is red only.

Table III. Red-channel forward voltage and series resistance by condition. Uncertainties are the standard deviation across channels.

Condition	Channels n	V_F at 10 mA (V)	R_s (Ω)	Ideality n	Reverse check
1	6	2.0021 ± 0.0105	10.77 ± 0.79	1.67–1.81	6 clean
2	5	2.0048 ± 0.0058	11.12 ± 0.34	1.66–1.80	5 clean
3	1	1.9989	10.92	1.65–1.65	1 clean, 1 shunted
4	3	2.0224 ± 0.0029	12.79 ± 0.65	1.69–1.81	3 clean
5	4	2.0109 ± 0.0129	11.28 ± 1.06	1.70–1.92	4 clean
6	3	2.0137 ± 0.0170	12.02 ± 1.20	1.67–1.80	3 clean
7	4	2.0132 ± 0.0170	11.46 ± 1.32	1.76–1.87	4 clean
8	3	2.0208 ± 0.0158	11.61 ± 0.92	1.75–2.06	3 clean

Across 29 red channels the mean series resistance is 11.43 Ω with a standard deviation of 0.99 Ω . One-way ANOVA across the eight conditions gives $F(7,21) = 1.73$, $p = 0.16$. Forward voltage at 10 mA behaves the same way: 2.0104 V mean, 12.9 mV standard deviation, $p = 0.24$. Neither quantity separates the assembly conditions.

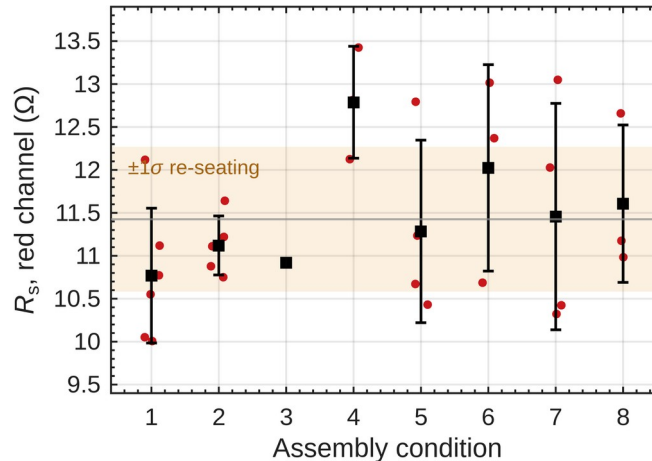


Fig. 4. Series resistance by condition. Points are individual channels, squares are condition means, bars are ± 1 standard deviation. The shaded band is the measurement repeatability established in Section D.

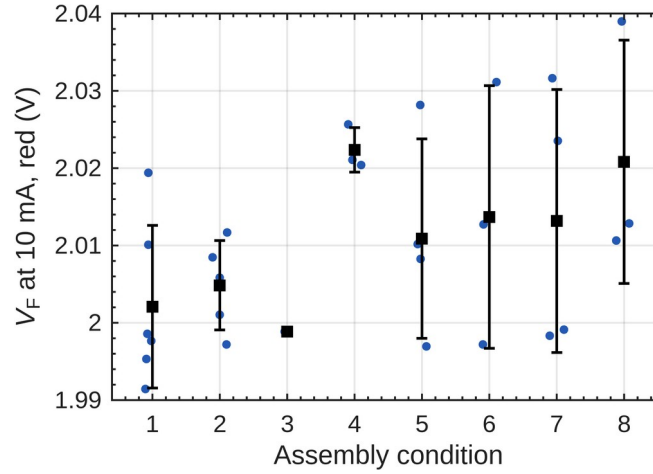


Fig. 5. Forward voltage at 10 mA by condition. The full spread across all 29 channels is 13 mV, which is consistent with dice drawn from one reel.

C. Defect detection from the sweep

One channel, S3 D1 red, fits to an ideality of 13.0 and a negative series resistance. Neither is physical. Below 3 mA its junction sits at 1.21 V, under a red LED's turn-on, while 0.42 mA still flows, so a parallel path of roughly 2.9 k Ω carries the current around the die. Above 3 mA the curve is indistinguishable from a healthy channel. A diode test at 1 mA reads 1.781 V and passes it, because at that current the shunt takes only a third of the current.

This makes the ideality factor a usable screen. Any fit outside 1.2 to 2.4 has something conducting in parallel with the junction, however normal the curve looks at working current. All 29 usable red channels fell inside that range; the one that did not sits on the coupon the screening had already flagged for solder contamination.

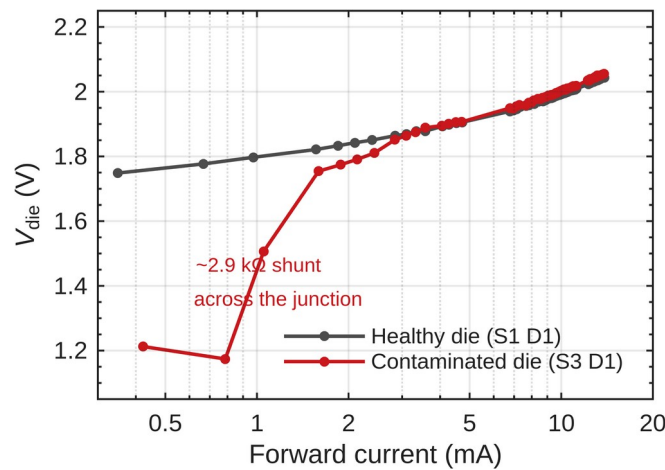


Fig. 6. A contaminated die (S3 D1) against a healthy one. The curves agree above 3 mA and diverge by 540 mV at the bottom of the sweep.

D. Measurement repeatability and detection limit

The series-resistance comparison is limited by the fixture, not by the dice. This build takes both voltage taps from the breadboard rather than from the Tier-1 probe pads, so the two female-male jumpers and their header contacts fall inside the measured loop.

Three channels were measured three times each, unplugging and re-seating the same two jumpers between measurements and changing nothing else. The pooled standard deviation is $0.84\ \Omega$, against a within-condition standard deviation of $0.92\ \Omega$. Re-seating therefore accounts for about 85 % of the variance that would otherwise be read as die-to-die spread.

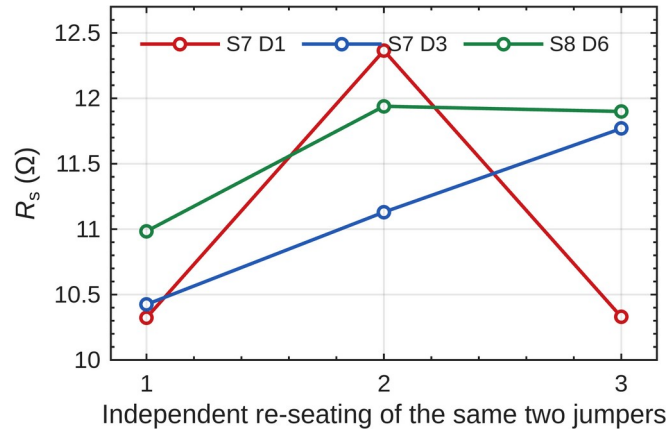


Fig. 7. Three channels, each measured after three independent re-seatings of the two jumpers. Nothing else changed between points.

Every re-seating lands at or above that channel's minimum, since contact resistance can only add. Taking the minimum of each channel gives 10.32 , 10.42 and $10.98\ \Omega$, so the true red series resistance is near $10.6\ \Omega$ and about $0.85\ \Omega$ of the reported mean is wiring.

Bond resistances are 10 to $100\ \text{m}\Omega$. At $n = 4$ the smallest difference in mean R_s this arrangement can resolve is $1.4\ \Omega$, more than an order of magnitude above the effect. The null result in Section B is therefore a statement about the instrument, not about the bonds. Moving the sense connections to the Tier-1 probe pads would put the jumper contacts outside the measured loop and recover the method.

E. Self-heating

The die dissipates $28.4\ \text{mW}$ at $13.9\ \text{mA}$. Junction heating lowers the forward voltage at high current, flattens the I - V curve and biases R_s downwards, so the pulse length has to be chosen rather than assumed. One channel was swept three times with only the integration length changed, giving current-on times of 5 , 20 and $80\ \text{ms}$ per point.

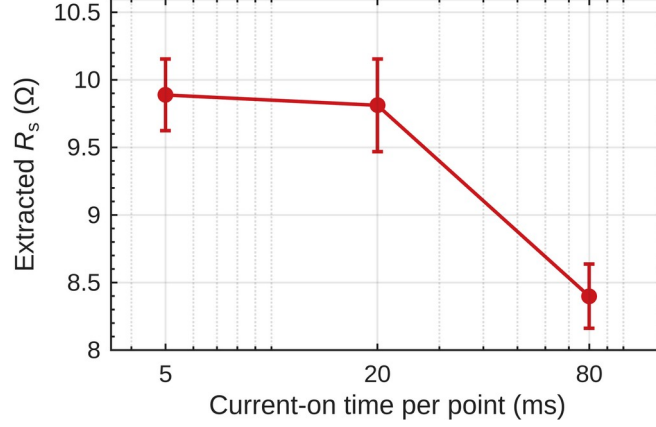


Fig. 8. Extracted series resistance against current-on time, same channel throughout. Error bars are the standard error of the fit.

Between 5 and 20 ms the result is unchanged within the fit error. At 80 ms it falls by 1.49Ω . For a junction-to-board thermal resistance of order 200 K/W, 28.4 mW gives about 6 K of rise, and at -2 mV/K that appears as roughly -1.2Ω of apparent series resistance, matching the measurement to within 25 %. The package thermal time constant therefore lies between 20 and 80 ms. All campaign data was taken at 5 ms.

F. Reverse bias check

Every surviving red channel was also biased in reverse to look for conduction paths around the junction. The die sits in series with the $100 \text{ k}\Omega$ across the 5 V rail and the meter reads the voltage across the die, so the die's own resistance sets the reading. A healthy junction leaves the meter's $10 \text{ M}\Omega$ input impedance as the only load and reads 4.99 V; a shunt pulls it down. Reading the small voltage across the sense resistor instead does not work with a handheld meter, because 1 nA through $98 \text{ k}\Omega$ is 0.1 mV and sits inside the meter's noise.

Thirty-four channels were measured. Thirty read 4.988 to 4.989 V, the open-die limit, which places their shunt resistance above $10 \text{ M}\Omega$ and their leakage below roughly 50 nA. Three read 0 V and confirm the round 1 shorts. One channel, S3 D1 red, read 0.085 V, which corresponds to a shunt of about $1.7 \text{ k}\Omega$.

That channel is the one the sweep had already flagged in Section C, where the forward characteristic implied roughly $2.9 \text{ k}\Omega$. Two measurements at opposite bias polarity agree to within a factor of two on a defect that a 1 mA diode test passes. It was recorded as “suspect” in the round 1 screening on the operator's judgement alone; it is now confirmed, and condition 3's yield stands as measured rather than estimated.

The result is otherwise a null, and worth stating as one. Condition 8 loses two channels to shorts, but its survivors are indistinguishable from those of conditions 5 and 7, so its failures are isolated rather than a coupon-wide contamination problem. No condition other than 3 carries a partial conduction path. Reverse bias corroborates the yield ranking; it does not add a new axis to it.

G. Summary

Table IV. Summary across assembly conditions.

Condition	Yield (%)	Failure modes	V_F at 10 mA (V)	R_s (Ω)	Reverse	Assessment
1	83.3	open 1, detached 3	2.002	10.77	clean	under-bonded
2	58.3	open 4, detached 6	2.005	11.12	clean	under-bonded
3	25.0	suspect 1, cross-lit 2, open 2, short 4	1.999	10.92	1 shunt	over-bonded
4	66.7	suspect 1, cross-lit 1, open 1, short 1	2.022	12.79	clean	over-bonded
5	100.0	none	2.011	11.28	clean	in window
6	75.0	detached 3	2.014	12.02	clean	under-bonded
7	100.0	none	2.013	11.46	clean	in window
8	83.3	short 2	2.021	11.61	clean	over-bonded

Three results stand. Channel yield and failure mode separate the eight conditions at $p = 2.2 \times 10^{-4}$ and place conditions 5 and 7 ahead of the rest, with no failures in 12 channels each. Series resistance does not separate them, and the reason is measured rather than assumed: fixture repeatability of 0.84Ω exceeds the bond resistances by more than an order of magnitude. Reverse bias found one partial shunt in 34 channels and confirms the yield ranking rather than adding to it.

The failure modes give a direction as well as a ranking. Conditions 1, 2 and 6 lose channels to detachment and opens, so they are under-bonded. Conditions 3, 4 and 8 lose them to shorts and bridges, so they are over-bonded. Conditions 5 and 7 lose none. The reverse check adds that the surviving joints are electrically sound on every condition except 3, which narrows the gap between condition 8 and the two clean conditions: its failures are isolated shorts rather than a coupon-wide problem.

The one measurement still worth recovering is series resistance. Sensing at the Tier-1 probe pads instead of the breadboard would move the jumper contacts outside the measured loop and bring the repeatability from 0.84Ω down to the fit precision, which was 0.16Ω once the firmware was corrected. That is the change that would let R_s resolve a bond difference.

Not measured: TLM ladders, van der Pauw cloverleaves and impedance spectroscopy, all of which need a four-wire source-measure unit or an LCR meter. The DCL6 and DCL12 daisy chains read open on every coupon because the chain routes each die's top-left pad to its top-right pad; with the dice rotated 90° those are the red and blue cathodes, so every element is a pair of back-to-back diodes and the anode is left floating. That is a layout fault in this coupon revision, not a bonding failure.